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Technical Report

Iris Flower Classification Using Machine Learning

OIB-SIP Internship – Task 1 (Machine Learning)

Prepared using Python, scikit-learn, pandas, and seaborn

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1. Executive Summary

This project builds and compares four supervised machine learning classifiers – Logistic Regression, K-Nearest Neighbours, Decision Tree, and Random Forest – to identify the species of an iris flower (Setosa, Versicolor, or Virginica) from four physical measurements. The dataset, sourced from scikit-learn, contains 150 samples split evenly across the three species with no missing values. Exploratory analysis showed that petal length and petal width are far more discriminative than the sepal measurements, a finding later confirmed quantitatively by Random Forest feature importance scores.

After stratified 80/20 train-test splitting and feature standardisation, all four classifiers achieved an identical test accuracy of 93.33% (28 of 30 samples correct), with every misclassification occurring between Versicolor and Virginica – never involving Setosa. Five-fold cross-validation was used to break the tie: Random Forest showed the most stable performance across folds (lowest standard deviation), and was therefore selected as the recommended model, with Logistic Regression noted as a strong, simpler alternative. The sections that follow document the dataset, the analysis pipeline, the modelling approach, and the full set of evaluation results that support these conclusions.

2. Introduction

2.1 Background

Flower species classification is a classic entry point into supervised machine learning because it is simple enough to reason about by hand yet rich enough to demonstrate a full modelling pipeline. The Iris dataset used in this project was introduced by the statistician and biologist Ronald Fisher in 1936 as an example of linear discriminant analysis, and has since become one of the most widely used

benchmark datasets in pattern recognition and introductory machine learning education. Its popularity comes from a combination of small size, clean structure, and a well-understood ground truth, which makes it ideal for comparing how different classification algorithms behave under the same conditions.

2.2 Objectives

The goal of this task was to move beyond simply calling a classifier's fit and predict methods, and instead to follow a complete and defensible data science workflow. Specifically, the project set out to:

- Load and inspect the dataset to confirm its integrity (shape, data types, missing values, class balance).

- Perform exploratory data analysis to understand feature distributions and class separability. ● Reason about which features are likely to be useful for classification, and confirm this with a quantitative feature importance measure.
- Correctly prepare the data for modelling, including a stratified split and feature scaling that avoids data leakage.
- Train multiple classification algorithms of different types (linear, distance-based, tree-based, ensemble) on identical data.
- Evaluate and compare the models using more than one metric, including cross-validation, rather than relying on a single train/test accuracy figure.
- Select and justify a best-performing model based on the full body of evidence collected.

2.3 Scope of This Report

This report documents the notebook *iris_classification.ipynb* end to end. Every number reported here was produced by executing that notebook's exact logic and parameters, including its fixed random seed (`random_state = 42`), so the results in this document are reproducible from the accompanying code. The report is organised to mirror the notebook's structure: dataset description, exploratory analysis, feature selection, methodology, results, and conclusions.

3. Dataset Description

The classic Iris dataset was loaded directly from scikit-learn's built-in datasets module, so no external file download or manual cleaning was required. Each of the 150 samples represents a single flower and is described by four continuous numeric measurements, all in centimetres, together with a species label.

Property	Value
Total samples	150
Features	4 (sepal length, sepal width, petal length, petal width)
Target classes	3 (Setosa, Versicolor, Virginica)
Samples per class	50 (perfectly balanced)
Missing values	0 (no imputation required)
Feature data type	Continuous, floating point, measured in centimetres

Table 1: Summary of dataset properties.

Because the dataset is perfectly balanced across the three species, accuracy is a fair and unbiased headline metric here – a model cannot inflate its score simply by favouring a majority class, since no

majority class exists. This also removes the need for resampling techniques or class-weighting adjustments that are often necessary with imbalanced data.

3.1 Descriptive Statistics

Table 2 summarises the central tendency and spread of each feature across all 150 samples. Petal length shows by far the largest standard deviation and range of any feature, which is an early hint that it will carry strong discriminative power between species.

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
Sepal length (cm)	5.84	0.83	4.30	5.10	5.80	6.40	7.90
Sepal width (cm)	3.06	0.44	2.00	2.80	3.00	3.30	4.40
Petal length (cm)	3.76	1.77	1.00	1.60	4.35	5.10	6.90
Petal width (cm)	1.20	0.76	0.10	0.30	1.30	1.80	2.50

Table 2: Descriptive statistics of the four features across all 150 samples.

3.2 Feature Means by Species

Grouping the data by species, shown in Table 3, reveals a clear ordering: Setosa has the smallest petal measurements by a wide margin, Virginica the largest, and Versicolor sits consistently in between. Sepal width behaves differently from the other three features: Setosa actually has the largest average sepal width, despite having the smallest petals, which illustrates why sepal width alone is a weak predictor of species.

Species	Sepal Length	Sepal Width	Petal Length	Petal Width
Setosa	5.01	3.43	1.46	0.25
Versicolor	5.94	2.77	4.26	1.33
Virginica	6.59	2.97	5.55	2.03

Table 3: Mean feature values grouped by species.

4. Exploratory Data Analysis

Before any model was trained, the dataset was examined visually and statistically to build an understanding of feature relationships and class separability. Four complementary visualisations were produced in the notebook, each generated as a saved figure (pairplot.png, boxplots.png, heatmap.png, violins.png).

4.1 Pair Plot

A pair plot was generated showing every pairwise scatter plot between the four features, coloured by species, with kernel density estimates on the diagonal. This single figure gives the clearest overall picture of separability: Setosa forms a tight, completely isolated cluster in almost every feature pair, particularly those involving petal length or petal width. Versicolor and Virginica form two more elongated, adjacent clusters with a visible but modest region of overlap, mostly in the sepal-based scatter plots.

4.2 Box Plots

Box plots were produced for each of the four features, grouped by species, arranged in a 2x2 grid. These confirm the pattern seen in the pair plot in a more quantitative form: the interquartile ranges for petal length and petal width barely overlap between Setosa and the other two species, and only partially overlap between Versicolor and Virginica. Sepal length and especially sepal width show much greater overlap across all three species, reinforcing that they carry comparatively weak discriminative signal on their own.

4.3 Correlation Heatmap

A correlation heatmap of the four numeric features was computed to check for strong linear relationships. Petal length and petal width are strongly positively correlated with each other, and both are also strongly correlated with sepal length. Sepal width stands out again as the odd feature out, showing weak or slightly negative correlation with the other three. This correlation structure is consistent with the biological intuition that overall flower size (captured jointly by petal length, petal width, and sepal length) scales together across the three species, while sepal width varies more independently.

4.4 Violin Plots

Violin plots were produced for the two most informative features – petal length and petal width – since a violin plot combines a box plot with a kernel density estimate and therefore reveals the shape of each species' distribution, not just its summary statistics. Both plots show three visually distinct, mostly non-overlapping density regions, one per species, which visually explains why classifiers built primarily on these two features are able to reach high accuracy.

4.5 Summary of EDA Insights

- Setosa is linearly separable from the other two species on almost every feature, with petal length and petal width showing near-zero overlap.
- Versicolor and Virginica overlap more than Setosa does with either, particularly in sepal measurements, but remain largely distinguishable using petal dimensions.
- Petal length and petal width are strongly correlated with each other and are the most discriminative pair of features.
- Sepal width is the weakest and most independent feature, and is expected to contribute the least to any trained classifier.

5. Feature Selection and Importance

Not all features contribute equally to the classification task. Based on the visualisations above and general domain understanding of iris morphology, each feature's usefulness can be assessed qualitatively before any model is trained, as summarised in Table 4.

Feature	Discriminative Ability	Reasoning
Petal length	High	Nearly zero overlap between all three species
Petal width	High	Tight, well-separated clusters per species
Sepal length	Moderate	Setosa separates well; Versicolor/Virginica overlap
Sepal width	Low	Heavy overlap across all three species

Table 4: Qualitative feature assessment based on exploratory analysis.

To turn this qualitative judgement into a quantitative measure, a Random Forest classifier was fitted on the full dataset purely to extract feature importance scores. This step is for analysis only: the actual train/test evaluation used in Section 7 uses a separate held-out split, so this preliminary fit does not leak information into the final results.

Feature	Importance Score
Petal length (cm)	0.436
Petal width (cm)	0.436
Sepal length (cm)	0.106
Sepal width (cm)	0.022

Table 5: Random Forest feature importance scores (full-dataset fit).

The quantitative result matches the qualitative assessment closely: petal length and petal width together account for roughly 87% of the model's decision-making signal, sepal length contributes just over 10%, and sepal width contributes barely 2%. No features were removed before training the final models, since even weak features can still provide marginal benefit and none of the four algorithms used here require manual dimensionality reduction on a feature set this small.

6. Methodology

6.1 Data Preparation

The feature matrix and target vector were separated, and the data was split into training (80%, 120 samples) and testing (20%, 30 samples) subsets using scikit-learn's `train_test_split` with `stratify` set to the target labels. Stratification guarantees that each of the three species is represented in the same proportion in both the training and test sets (10 samples per class in the test set), which is important given how small the dataset is: without it, a single unlucky split could leave a class under-represented in the test set and make the resulting accuracy figure unreliable.

Feature values were then standardised using `StandardScaler`, which transforms each feature to have zero mean and unit variance:

$$z = (x - \mu) / \sigma$$

where μ and σ are the mean and standard deviation of a feature computed on the training set only. The same transformation (using the training set's μ and σ) was then applied to the test set. Fitting the scaler on the training data alone and re-using it for the test data, rather than fitting it separately on each set, avoids data leakage: it ensures that no information from the test set influences how the training data is scaled. Feature scaling matters most for K-Nearest Neighbours, which relies directly on distances between points, and for Logistic Regression, whose gradient-based optimiser converges more reliably on standardised inputs. Decision Tree and Random Forest are scale-invariant by construction, but were trained on the same scaled data for consistency across all four models.

6.2 Classification Algorithms

Four classifiers, representing four different modelling philosophies, were trained on the identical scaled training data so that their results are directly comparable.

6.2.1 Logistic Regression

A linear model that estimates the probability of each class using the logistic (softmax, in the multi-class case) function applied to a weighted sum of the input features. It was configured with `max_iter = 200` to

ensure the optimiser converges, and `random_state = 42` for reproducibility. Logistic Regression serves as an interpretable baseline: its coefficients directly indicate how strongly each feature pushes the prediction toward or away from a given class.

6.2.2 K-Nearest Neighbours (KNN)

A non-parametric, instance-based method that classifies a new sample by majority vote among its *k* nearest neighbours in the training set, measured using Euclidean distance. It was configured with `n_neighbors = 5`. KNN makes no assumption about the shape of the decision boundary, which makes it flexible, but its predictions depend heavily on having features on comparable scales, which is why standardisation is especially important here.

6.2.3 Decision Tree

A rule-based model that recursively splits the feature space into regions that are as pure (single-class) as possible, using a greedy criterion at each split. It was configured with `max_depth = 4` to limit complexity and reduce the risk of overfitting on a dataset this small. A key advantage of a Decision Tree is that its decision logic can be read directly as a sequence of if/else rules on the input features.

6.2.4 Random Forest

An ensemble method that trains many individual decision trees (`n_estimators = 100`, `max_depth = 5`) on bootstrapped subsets of the training data and random subsets of features, then combines their predictions by majority vote. Averaging over many trees reduces the variance that a single Decision Tree is prone to, generally producing a more stable and better-generalising model at the cost of reduced interpretability of any single decision path. This is also the model used in Section 5 to compute feature importance scores.

6.3 Evaluation Metrics

Each model was evaluated using a combination of metrics rather than a single number, since accuracy alone can hide class-specific weaknesses. The metrics used, and their formulas in terms of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), are as follows:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Because this is a three-class problem, precision, recall, and F1-score were computed per class and then macro-averaged (a simple, unweighted average across the three classes), which treats each species as equally important regardless of how many test samples it has. In addition, a full confusion matrix was produced for every model, showing exactly which species were confused with which, and a 5-fold cross-validation score was computed on the training set to test how consistently each model performs across different subsets of the data, independent of the one specific train/test split used for the headline accuracy figure.

7. Results

7.1 Overall Model Comparison

Model	Test Acc.	CV Mean	CV Std	Precisio n	Recal l	F1- score
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Logistic Regression	93.33 %	95.83%	0.026	0.933	0.933	0.933
K-Nearest Neighbours	93.33 %	96.67%	0.031	0.944	0.933	0.933
Decision Tree	93.33 %	94.17%	0.020	0.933	0.933	0.933
Random Forest	93.33 %	95.00%	0.017	0.933	0.933	0.933

Table 6: Test-set accuracy, macro-averaged precision/recall/F1, and 5-fold cross-validation results for all four models. CV Mean and CV Std are computed on the training set only.

All four models reach an identical test accuracy of 93.33% (28 of 30 test samples correctly classified), so the test set alone does not separate them. Cross-validation results, which average performance over five different training/validation splits, are therefore used as a tie-breaker. K-Nearest Neighbours obtains the highest mean CV accuracy (96.67%) but also the highest variability (standard deviation = 0.031). Random Forest has a slightly lower CV mean (95.00%) but the lowest standard deviation (0.017) of all four models, indicating the most consistent performance across folds. Decision Tree has the lowest CV mean (94.17%), which is consistent with individual decision trees being more prone to variance than an ensemble.

7.2 Per-Model Classification Reports

Tables 7-10 give the full per-class precision, recall, and F1-score for each model on the 30-sample test set, with 10 true samples per class.

Logistic Regression

Species	Precision	Recall	F1-score	Support
Setosa				10
Versicolor	1.00	1.00	1.00	10
Virginica	0.90	0.90	0.90	10
	0.90	0.90	0.90	

Table 7: Logistic Regression per-class results.

K-Nearest Neighbours

Species	Precision	Recall	F1-score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.83	1.00	0.91	10
Virginica	1.00	0.80	0.89	10

Table 8: K-Nearest Neighbours per-class results.

Decision Tree

Species	Precision	Recall	F1-score	Support
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Setosa				10
Versicolor	1.00	1.00	1.00	10
Virginica	0.90	0.90	0.90	10
	0.90	0.90	0.90	

Table 9: Decision Tree per-class results.

Random Forest

Species	Precision	Recall	F1-score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.90	0.90	0.90	10
Virginica	0.90	0.90	0.90	10

Table 10: Random Forest per-class results.

7.3 Confusion Matrices

Tables 11-14 show the full 3x3 confusion matrix for each model on the 30-sample test set. Rows represent the true species and columns represent the predicted species, so the diagonal entries are correct predictions and off-diagonal entries are errors.

Logistic Regression

True \ Predicted	Setosa	Versicolor	Virginica
Setosa			0
Versicolor	10	0	1
Virginica	0	9	9
	0	1	

Table 11: Logistic Regression confusion matrix.

K-Nearest

Neighbours

True \ Predicted	Setosa	Versicolor	Virginica
Setosa			0
Versicolor	10	0	0
Virginica	0	10	8
	0	2	

Table 12: K-Nearest Neighbours confusion matrix.

Decision Tree

True \ Predicted	Setosa	Versicolor	Virginica
Setosa	10	0	0
Versicolor	0	9	1
Virginica	0	1	9

Table 13: Decision Tree confusion matrix.

Random Forest

True \ Predicted	Setosa	Versicolor	Virginica
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Setosa	10	0	0
Versicolor	0	9	1
Virginica	0	1	9

Table 14: Random Forest confusion matrix.

Every model classifies all 10 Setosa test samples correctly, with zero exceptions across all four confusion matrices. The only errors, for every model, involve confusion between Versicolor and Virginica: Logistic Regression, Decision Tree, and Random Forest each misclassify exactly one Versicolor sample as Virginica and one Virginica sample as Versicolor, while K-Nearest Neighbours instead classifies all 10 Versicolor samples correctly but misclassifies 2 Virginica samples as Versicolor. This is consistent with the partial feature overlap between these two species observed throughout the exploratory analysis in Section 4.

8. Discussion

The results confirm that the Iris dataset is close to linearly separable, and that even a simple linear model such as Logistic Regression performs on par with more flexible non-linear models. This is expected given the small size and clean structure of the dataset: with only 150 samples and 4 well-behaved numeric features, all four algorithms have enough capacity to capture the underlying class boundaries, and the ceiling on achievable accuracy is set by the genuine overlap between Versicolor and Virginica rather than by model choice.

The feature importance analysis in Section 5 and the confusion matrices in Section 7.3 agree on this point from two independent angles: virtually all misclassification happens between Versicolor and Virginica, never involving Setosa, and petal measurements – the features that best separate Setosa from the others – are consistently the dominant signal in every model. Where the models genuinely differ is not in their test accuracy but in how they resolve the ambiguous Versicolor/Virginica boundary and in how consistently they do so across different random splits of the data, which is exactly what the cross-validation results in Table 6 are designed to expose.

K-Nearest Neighbours' behaviour is also worth noting: because it makes predictions based on local neighbourhoods rather than a single global decision boundary, its error pattern (all Versicolor correct, 2 Virginica samples pulled toward Versicolor) is qualitatively different from the other three models, even though its overall test accuracy is the same. This illustrates why relying on a single aggregate accuracy number can be misleading when comparing models – the confusion matrix reveals a difference in behaviour that accuracy alone does not.

9. Conclusion and Best Model

Random Forest is selected as the best-performing model for this task. The justification is based on the following combination of evidence gathered across Sections 5 through 8:

- Test accuracy: It matches the highest test accuracy achieved by any model (93.33%), correctly classifying 28 of 30 held-out samples.
- Cross-validation stability: It has the lowest cross-validation standard deviation (0.017) of all four models, meaning its performance varies the least across different data splits – a sign that it generalises reliably rather than fitting one particular split.

- **Robustness to weak features:** As an ensemble of decision trees built on random feature subsets, Random Forest is naturally robust to the fact that sepal width contributes very little signal, without requiring manual feature elimination.
- **Interpretability at the ensemble level:** It provides feature importance scores (Table 5), which give a transparent, quantitative explanation of its decisions (petal dimensions dominate) rather than acting as a pure black box.

Logistic Regression remains a strong, lightweight alternative given its equal test accuracy, simplicity, and direct interpretability through its learned coefficients, and would be a reasonable choice in a setting where model simplicity and low computational cost are prioritised over the marginally greater cross-validation stability of an ensemble method. Decision Tree, while easy to interpret as a single set of rules, showed the least stable cross-validation performance of the four and is not recommended as the primary model on its own, though it remains useful as a quick, explainable baseline.

10. Limitations and Future Work

While the results are consistent and well-supported by multiple evaluation angles, several limitations are worth noting, along with directions that could extend this work:

- **Dataset size:** With only 150 samples and a 30-sample test set, each test sample represents roughly 3.3% of the reported test accuracy, so small differences between models (a single misclassified sample) can look larger in percentage terms than they really are.
- **Single train/test split:** Although cross-validation was used to check stability, the headline test accuracy still comes from one specific 80/20 split; repeating the entire pipeline over multiple random seeds would give a more robust final estimate.
- **No hyperparameter search:** Hyperparameters (KNN's neighbour count, tree depths, forest size) were set to reasonable fixed values rather than tuned via grid search or cross-validated search, which could potentially improve or further separate model performance.
- **Limited feature set:** The analysis is restricted to the four classic measurements included in the dataset; incorporating additional features (if available) or engineered feature combinations could help resolve the remaining Versicolor/Virginica ambiguity.
- **Generalisation beyond this dataset:** Because the Iris dataset is small, clean, and well-studied, these results should not be assumed to transfer directly to noisier, larger, or less balanced real-world classification problems without re-validation.

Future extensions of this project could include a systematic hyperparameter search (for example, GridSearchCV) for each model, testing additional algorithms such as Support Vector Machines or Gradient Boosting, and visualising decision boundaries in two dimensions using the two most important features to give an intuitive, geometric picture of how each model resolves the Versicolor/Virginica boundary.

11. Tools and Technologies

Component	Tool / Library
Language	Python 3.10+
Data handling	pandas, numpy

Machine learning	scikit-learn
Visualisation	matplotlib, seaborn
	Jupyter
Development environment	Notebook

Table 15: Technology stack used in this project.

12. Project Deliverables

The complete project consists of the notebook itself together with a set of generated figures and supporting files, summarised in Table 16.

File	Description
iris_classification.ipynb	Main notebook containing all code, analysis, and narrative
requirements.txt	Python package dependencies
README.md	Project overview, setup instructions, and checklist
pairplot.png	Pairwise feature scatter plot, coloured by species
boxplots.png	Per-feature box plots grouped by species
heatmap.png	Feature correlation heatmap
violins.png	Violin plots of petal length and petal width
feature_importance.png	Random Forest feature importance bar chart
confusion_matrices.png	Confusion matrices for all four models
accuracy_comparison.png	Bar chart comparing test accuracy across models

Table 16: Project files and generated outputs.

Note: All figures and numeric results in this report were reproduced by re-executing the project notebook’s logic with the same random seed (random_state = 42) and the same hyperparameters as defined in iris_classification.ipynb.